QUANTUM MECHANICS

Quantum Mechanics (Wave –Mechanics)

In classical mechanics the motion of particle is usually described using the time dependent position x(t) as the dynamical variable. For example F = ma = mẍ.

In wave mechanics the dynamical variable is a wave function.

The wave function depends on position and time and it is a complex number, i.e. ψ (**r**, t) $\in \mathbb{C}$ (pronounced as ψ (psi) belongs to complex).

Quantum mechanics was developed in 1925 by Schrödinger and Heisenberg. Schrödinger method involves partial difference equation. Heisenberg's method employs matrices. Since former involves the solution of wave equations, it is also called **wave-mechanics**; and latter, which involves matrices, is called **matrix mechanics**. Though both of them are equivalent.

In this syllabus we shall mostly confine ourselves to wave-mechanics only.

The Schrödinger Equation

If a mass (particle) 'm' is moving along x-axis subject to a specified force F(x,t). Classical mechanics looks to determine the position of the particle at any given time x(t). Once we know this information v, p etc. can be determined. Relevant equation is Newton's 2nd law of motion

$$F = ma$$

or

$$m\frac{d^2x}{dt^2} = -\frac{\partial v}{\partial x}$$

Where, we have assumed i) force is not magnetic and ii) motion is not relativistic.

Quantum mechanics approaches the problem quite differently. Here we look for a wave function, ψ (x, t), of the particle and we get it by solving the **Schrödinger Equation**

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

The corresponding 3 dimensional equation Schrödinger Equation is

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi\right]$$

Here, $i = \sqrt{-1}$ and $\hbar = \frac{h}{2\pi}$ = reduced Planck's constant $\approx 1.054 \times 10^{-34}$ J-s.

Given suitable initial condition ψ (x, 0), we can determine ψ (x, t) for all future time, as in classical mechanics Newton's law determines x(t) for all future time.

There are several ways to develop wave equation by applying quantum concepts to various classical equations of mechanics. One is to consider a few basic postulates, develop the wave equation from them and rely on the accuracy of the results to serve as a justification of the postulates.

Basic postulates

1. Each particle in a physical system is described by a wave function ψ (x, y, z, t). This function and its space derivative $(\frac{\partial \Psi}{\partial x} + \frac{\partial \Psi}{\partial y} + \frac{\partial \Psi}{\partial z})$ are continuous, finite and single valued.
   N.B.
   i) ψ is related to a physical quantity, so it can't be discontinuous at any point.
   ii) If ψ is infinite it would imply infinite large probability of finding the particle ín that region, which is in violation of uncertainty principle.
   iii) Single valued as the function $\int_{-\infty}^{\infty} \Psi^* \Psi \, dx$ must have oly one numerical value.
2. In dealing with classical quantities such as energy E and momentum p, we must relate these quantities with abstract quantum mechanical operators as following

Classical variables	Quantum operator
x	x
f(x)	f(x)
p(x)	$-i\hbar \frac{\partial}{\partial x}$
E	$i\hbar \frac{\partial}{\partial t}$
V(x)	V(x)

   And similarly for other directions.

3. The **statistical interpretation of the wave function**
   By nature particle is localized, whereas the wave function is spread out in space. How can such an object describe the state of a particle? To understand this we need to answer the following questions. What is a wave function? What is it significance? What we can do with it?
   The answer to all of the above questions is given by Bohr's statistical interpretation of the wave function:

$|\Psi(x,t)|^2$ gives the probability of finding the particle at point x at time t. i.e.

$|\Psi(x,t)|^2$ dx = probability of finding the particle between x and (x + dx) at time t.

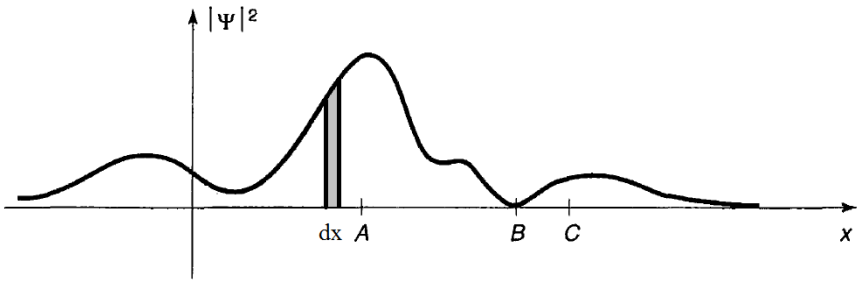


Fig. 1. A typical wave function.

The particle would be relatively likely to be found near A, unlikely to be near B. The shaded area represents the probability of finding the particle in the range dx.

The statistical interpretation introduces a kind of indeterminacy into quantum mechanics, i.e you can't predict with certainty the outcome of a simple experiment like to measure the position of a particle. **Quantum mechanics offers statistical information about possible results**.

Is this a peculiarity of the nature, a deficiency in the theory, a fault in the measurement apparatus, or what?

If I measure the position of a particle and I find it to be at point C. Where was the particle just before I made the measurement?

1. The realist position: It was at C. (Einstein believed this). If this is true then quantum mechanics is an incomplete theory since the particle was at C and quantum mechanics was unable to tell us so. (In other words, indeterminacy is not a fact of nature, but a reflection of our ignorance).
2. The orthodox position (Copenhagen interpretation): The particle wasn't really anywhere. It was the act of measurement that forced the particle to take a stand.
3. The agnostic position: Refuse to answer.

In 1964 John Bell came up with an astonishing result that it makes observable difference if the particle had a precise (though unknown) position prior to the measurement. An immediate and repeated measurement must return the same value. It is said that the wave function collapsed upon measurement, to a spike at the point C (say). (ψ, however, will soon spread in accordance with the Schrödinger equation, so the second measurement must be made quickly).

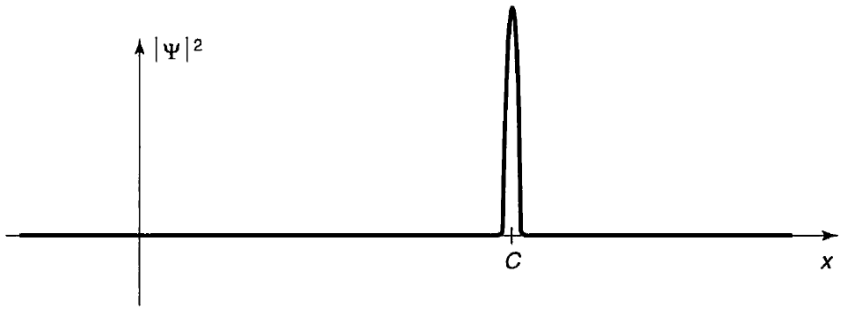


Fig. 2. Collapse of wave function immediately after the measurement found the particle at C.

Therefore, there are two distinct physical processes, (i) the ordinary one where wave function evolves leisurely as per Schrödinger equation, and (ii) sudden and discontinuous collapse of the eave function due to measurements.

Normalization of Wave function

The product $|\Psi(\mathrm{x},\mathrm{t})|^2 = \Psi^*\Psi$, Ψ^* being the complex conjugate of Ψ, is normalized according to

$$\int_{-\infty}^{\infty}|\Psi|^2\,dx = \int_{-\infty}^{\infty}\Psi^*\Psi\,dx = 1$$

And **the average or expectation value** $\langle\alpha\rangle$ of any variable α is calculated from the wave function by using the general formula

$$\langle\alpha\rangle = \frac{\int_{-\infty}^{\infty}\Psi^*(x)\alpha\Psi(\mathrm{x})dx}{\int_{-\infty}^{\infty}\Psi^*(x)\Psi(\mathrm{x})dx}$$

If, however, if the wave function is normalized then $\int_{-\infty}^{\infty}\Psi^*(x)\,\Psi(\mathrm{x})dx = 1$ and the expectation value is given by

$$\langle\alpha\rangle = \int_{-\infty}^{\infty}\Psi^*(x)\alpha\,\Psi(\mathrm{x})dx$$

Once we find the wave function Ψ for particle, we can calculate its average position, energy or momentum within the limits of uncertainty principle. So major effort in quantum mechanics is to solve Ψ within the conditions imposed by a particular physical system.

Formulation of Schrödinger Equation

The classical equation for energy of a particle of mass *m*

Kinetic Energy + Potential Energy = Total Energy

or $\frac{1}{2}mv^2 + V = E$

or $\frac{p^2}{2m} + V = E$ where $p = mv$ is the momentum of the particle.

Using quantum mechanical operators to for p (= - $i\hbar\frac{\partial}{\partial x}$) and E (= $i\hbar\frac{\partial}{\partial t}$) we get if we allow the operators to operate on the wave function $\psi(x,t)$

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x)\psi(x,t) = i\hbar\frac{\partial\psi(x,t)}{\partial t} \quad (1)$$

Eqn. 1 is the Schrödinger Equation for one dimensional problem. Here the operational interpretation of $(\frac{\partial}{\partial x})^2$ is the second derivative form for $\frac{\partial^2}{\partial x^2}$, and $i^2 = -1$.

In three dimension the Schrödinger Equation is

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t} \quad (2)$$

If we assume potential, V, is independent of time, t, then we can solve the Schrödinger Equation by the method of separation of variables. We look for solution that are simple product

$$\Psi(x,t) = \psi(x)f(t) \quad (3)$$

Using Eqn. 3 in Eqn. 1 we get

$$-\frac{\hbar^2}{2m}f(t)\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x)f(t) = i\hbar\psi(x)\frac{\partial f(t)}{\partial t}$$

Dividing both sides by $\psi(x)f(t)$ gives

$$-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + V(x) = i\hbar\frac{1}{f(t)}\frac{df(t)}{dt} \quad (4)$$

Since left hand side is a function of 'x' only and right hand side is a function of 't' alone, this is true only if both sides are separately equal to a constant. Let the constant be E.

Now the variables can be separated and we get time dependent Schrödinger Equation

$$\frac{df(t)}{dt} = -\frac{iE}{\hbar}f(t)$$

Or $$\frac{df(t)}{dt} + \frac{iE}{\hbar}f(t) = 0 \quad (5)$$

And the time independent Schrödinger Equation

$$-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + V(x) = E$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \quad (6)$$

Eqn. (5) and (6) are time dependent and time independent Schrödinger Equation, respectively. Thus the separation of variables has turned a differential equation into two ordinary differential equations.

The general solution of Eqn. (5) is $C.\exp(-iEt/\hbar)$. But we may absorb the constant C into ψ, as our quantity of interest is $\psi(x)f(t)$. Then

$$f(t) = e^{\frac{-iEt}{\hbar}} \quad (7)$$

The Eqn. (6), known as time independent Schrödinger Equation, can be solved when the potential V is specified.

The general solution of $\Psi(x,t)$ is

$$\Psi(x,t) = \psi(x)e^{\frac{-iEt}{\hbar}} \quad (8)$$

Properties of Schrödinger's equation

1. Linearity: The Schrödinger equation is linear in the wave-function Ψ which means that the equation does not involve higher powers of the wave-function or its derivatives.

2. Superposition: The linearity of Schrödinger's equation implies that a linear combination of solutions of Schrödinger's equation for any system is also a solution. Hence, if Ψ_1 and Ψ_2 satisfy the Schrödinger's equation then

$$\Psi = a_1\Psi_1 + a_2\Psi_2 \quad (9)$$

is also a solution, where $a_1$and $a_2$are constants. Hence, the wave-functions Ψ_1 and $\Psi_2$obey the superposition principle. The superposition principle is the main difference between quantum mechanics and classical mechanics. The fact that particles can be described by waves whose amplitudes can be added cannot be explained by classical mechanics.

Operators, Eigenfunctions, Eigenvalues, Expectation values

Every physical observable (quantity) in Classical Mechanics can be represented by an operator in Quantum Mechanics which acts on the wave-function (a particular state of the particle) to produce another wave-function. Thus,

$$\hat{O}\Psi = \Psi' \tag{10}$$

Where $\hat{O}$ represents the operator. The following table represents the operators associated with various observable quantities.

Observable (Quantity)	Operator
Position, x	x
Linear momentum, p	$-i\hbar\frac{\partial}{\partial x}$
Potential Energy, U(x)	U(x)
Kinetic Energy, KE = $\frac{p^2}{2m}$	$-\frac{\hbar^2}{2m}\nabla^2$
Total Energy, E	$i\hbar\frac{\partial\Psi}{\partial t}$
Hamiltonian H = KE (Kinetic Energy) + PE (Potential Energy)	$-\frac{\hbar^2}{2m}\nabla^2 + V$

Two operators $\widehat{O_1}$ and $\widehat{O_2}$ are said to commute if

$$\widehat{O_1}\widehat{O_2}\Psi = \widehat{O_2}\widehat{O_1}\Psi \text{ or } \widehat{O_1}\widehat{O_2}\Psi - \widehat{O_2}\widehat{O_1}\Psi = 0 \tag{11}$$

for all Ψ. If two operators commute, they can be simultaneously determined precisely. The commutation relation for two operators $\widehat{O_1}$ and $\widehat{O_2}$ is given by

$$\left[\widehat{O_1}, \widehat{O_2}\right] = \widehat{O_1}\widehat{O_2} - \widehat{O_2}\widehat{O_1} \tag{12}$$

If two operators commute then $\left[\widehat{O_1}, \widehat{O_2}\right] = 0$

Example 2

Prove that the observables position x and momentum p_x cannot be determined simultaneously by using the commutation relation.

Solution:

$$[x, p_x]\,\Psi = (xp_x - p_x x)\,\Psi = -i\hbar\left(x\frac{\partial\Psi}{\partial x} - \frac{\partial}{\partial x}(x\Psi)\right) = -i\hbar\left(x\frac{\partial\Psi}{\partial x} - \left(x\frac{\partial\Psi}{\partial x} + \Psi\frac{\partial x}{\partial x}\right)\right)$$

$$\therefore\ [x, p_x]\,\Psi = i\hbar\Psi \qquad \text{or} \quad [x, p_x] = i\hbar$$

Since, the operators x and p_x do not commute, they cannot be determined simultaneously with 100% accuracies.

Given an operator $\hat{O}$, it is possible to find a wave-function $\Psi(x)$ such that

$$\hat{O}\,\Psi(x) = \lambda\Psi(x) \tag{13}$$

Where λ is a constant independent of x. The function $\Psi(x)$is called an eigenfunction of $\hat{O}$ and λ is the corresponding eigenvalue of $\hat{O}$.

N.B.

Eigenvalues

The eigenvalues are related to observed values in experimental measurements in the following manner. The observed value obtained from a single measurement is an eigenvalue.

Expectation values

In a large number of measurements (or measurement over a prolonged period of time), the measured value is an average value or expectation value. The expectation value of any observable quantity α is given by

$$\langle\alpha\rangle = \frac{\int_{-\infty}^{\infty}\Psi^*(x)\alpha\Psi(x)dx}{\int_{-\infty}^{\infty}\Psi^*(x)\Psi(x)dx} \tag{14}$$

Now, if the wave-function is normalized then $\int_{-\infty}^{\infty}\Psi^*(x)\,\Psi(x)dx = 1$ and the expectation value is given by

$$\langle\alpha\rangle = \int_{-\infty}^{\infty}\Psi^*(x)\alpha\,\Psi(x)dx \tag{15}$$

Some important properties of eigenvalues and eigenfunctions

1. If $\Psi(x)$ is an eigenfunction of an operator $\hat{O}$ with eigenvalue λ, then $b\Psi(x)$, where b is a constant, is also an eigenfunction of the same operator with the same eigenvalue λ. Hence, multiplying an eigenfunction with a constant does not change the eigenvalue.

2. If there are two eigenfunctions having the same eigenvalue, then the eigenfunctions are said to be degenerate eigenfunctions.

3. An operator is termed as a Hermitian operator if all its eigenvalues are real and the eigenfunctions corresponding to different eigenvalues are orthogonal.

Example 3

A particle limited to the x axis has wave-function $\Psi = ax$ between $x = 0$ and $x = 1$; while $\Psi = 0$ elsewhere, (a) Find the probability that the particle can be found between $x = 0.45$ and $x = 0.55$, (b) Find the expectation value of the particle's position.

Solution:

(a) The probability is $\int_{x_1}^{x_2} |\Psi|^2 dx = a^2 \int_{0.45}^{0.55} x^2 dx = a^2 \left[\frac{x^3}{3}\right]_{0.45}^{0.55} = 0.0251a^2$

(b) The expectation value is $\langle x \rangle = \int_0^1 x|\Psi|^2 dx = a^2 \int_0^1 x^3 dx = \frac{a^2}{4}$

Example 4

An eigenfunction of the operator d^2/dx^2 is $\psi = e^{2x}$. Find the corresponding eigenvalue.

Solution:

Let $\hat{O} = \frac{d^2}{dx^2}$, so, $\hat{O}\psi = \frac{d^2}{dx^2}(e^{2x}) = 4e^{2x} = 4\psi$

Comparing it with the Eq. $\hat{O}\,\Psi(x) = \lambda\Psi(x)$, we obtain $\lambda = 4$.

Hence, the eigenvalue of the given operator is 4.

Schrödinger's equation : Time-Dependent Form for a free particle

The one-dimensional time dependent Schrödinger equation for a free non-relativistic particle (V = 0) is given below

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} = i\hbar\frac{\partial\psi(x,t)}{\partial t} \quad (1)$$

A free particle of total energy E and momentum p moving in the +x direction can be described by the wave-function Ψ of a place monochromatic wave. The expression for Ψ is given by

$$\Psi(x,t) = A\exp\left(\frac{i}{\hbar}(px - Et)\right) \quad (2)$$

where A = amplitude of displacement of the wave

ħ = reduced Planck constant ($h/2\pi$) = 1.05457 x 10^{-34} J.s

The energy E and momentum p of the particle are related to its wave-vector k and frequency ω by the relations

$p = \hbar k = \hbar\frac{2\pi}{\lambda}$, $E = \hbar\omega$

Differentiating Eq. (2) once with respect to time gives,

$$\frac{\partial\Psi(x,t)}{\partial t} = -i\frac{E}{\hbar}\Psi(x,t)$$

or

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = E\Psi(x,t) \quad (3)$$

Differentiating Eq. (2) twice with respect to x gives,

$$-i\hbar\frac{\partial\Psi(x,t)}{\partial x} = p\Psi(x,t)$$

$$-\hbar^2\frac{\partial^2\Psi(x,t)}{\partial x^2} = p^2\Psi(x,t) \quad (4)$$

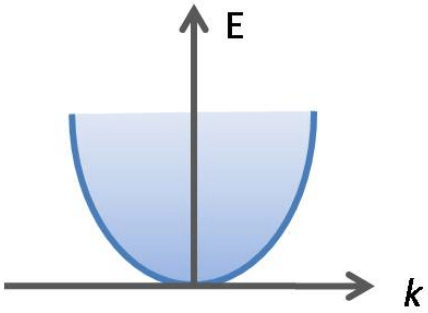
Dividing both sides of Eq. (4) by (1/2m), we obtain,

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} = \frac{p^2}{2m}\Psi(x,t) \quad (5)$$

Comparing equations (1), (3) and (5), the energy E and momentum p of a free non-relativistic particle of mass m are related by the expression

$$E = \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m} \quad (6)$$

As the particle is free, there are no boundary conditions to be applied to the particle motion and hence there are no restrictions on wave-vector k. As a result, the particle can possess any value of energy or the energy is not quantized. Thus, a freely moving particle possesses a continuous energy spectrum as given in the adjoining figure.

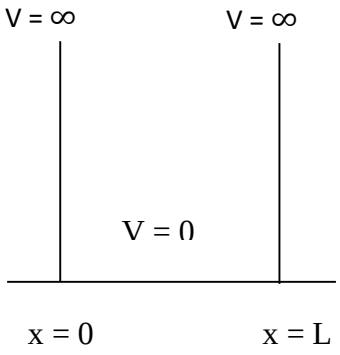


Some applications of Schrödinger's equation

Infinite Square-well Potential

Let us solve the problem of infinite potential well.

Let us assume that a particle of mass m trapped in a one-dimensional infinitely deep potential well with V(x) = 0 except at boundaries x=0 and x=L, where it is infinitely large as shown in the figure.



Therefore,

V(x) = 0 for 0 < x < L

= ∞ for x ≤ 0 and x ≥ L (1)

We have to find out the eigenfunction and the corresponding eigenvalues for this problem.

Inside the well V(x) = 0. Therefore, the time independent Schrödinger equation can be written as

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \qquad 0<x<L \qquad (2)$$

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0 \qquad (3)$$

Where $k^2 = \frac{2mE}{\hbar^2}$

This is simple harmonic oscillator equation. The general solution is

$$\psi(x) = A\sin kx + B\cos kx \qquad (4)$$

Where A and B are the arbitrary constants that can be found from boundary conditions.

Continuity of $\psi(x)$ requires

$$\psi(0) = \psi(L) = 0 \qquad (5) \quad \therefore$$

$$\psi(0) = A\sin(k.0) + B\cos(k.0) = 0$$

Therefore, $B = 0$

and $$\psi(x) = A\sin(kx) \qquad (6)$$

Now $$\psi(L) = A\sin(kL) = 0$$

Since A cannot be zero as it will give trivial, non normalizable solution $\psi(x) = 0$, we have

$$\sin(kL) = 0 \text{ or } kL = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots.$$

Again k=0 is not acceptable as it will imply $\psi(x) = 0$ and the '-'ve sign can be absorbed into A. Therefore we have

$$k_n = \frac{n\pi}{L} \qquad n = 1, 2, 3, ... \qquad (7)$$

From Eqn. (3) and (7), we solve for the total energy E_n for each value of the integer n

$$\frac{\sqrt{(2mE_n)}}{\hbar} = \frac{n}{L}$$

$$E_n = \frac{\pi^2 n^2 \hbar^2}{2mL^2} \qquad (8)$$

In contrast to the classical case, a quantum particle in the infinite square well can have only certain, discrete values of energy depending on the value of *n*. Thus, the energy becomes quantized due to confinement of the particle. Hence, a particle in an infinite potential well can have only bound states.

The energy E_1 is called the zero-point energy. Obviously, the energy is not *zero* in this case. The zero-point energy is the consequence of the uncertainty principle. If the energy of the particle is zero, its momentum is also zero. Then according to uncertainty principle, the position of the particle would become infinite and the particle cannot be confined to the box. Therefore, the particle cannot have zero energy.

The eigenfunctions corresponding to the above energy eigenvalues is given by

$$\psi_n = A \sin\ kx = A \sin \sqrt{\frac{2mE_n}{\hbar^2}}\, x \qquad (9)$$

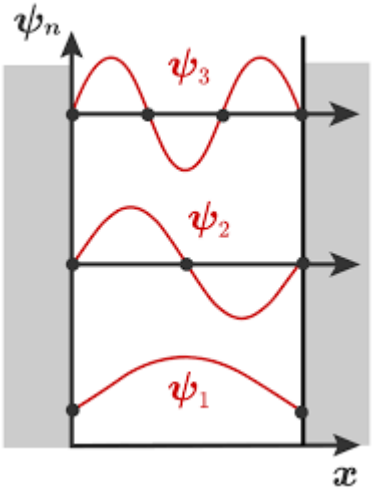
The constant A can be found from the normalization condition,

$$\int_0^L |\psi_n|^2 dx \ = \ A^2 \int_0^L sin^2 \left(\frac{n\pi x}{L}\right) dx = A^2 \frac{L}{2} = 1$$

Or $$A = \sqrt{\frac{2}{L}} \qquad (10)$$

Hence, the normalized wave-functions for a particle in a one-dimensional infinitely deep potential well is given by

$$\psi_n(x) = \sqrt{\frac{2}{L}}\, sin \frac{n\pi x}{L} \qquad \text{where n = 1, 2, 3, …….} \qquad (11)$$



First three wave functions are shown above.

Particle trapped in a three-dimensional box

For particle trapped in a three-dimensional square box of length a, the energy eigenvalues are given by

$$E = E_x + E_y + E_z = \left(n_x^2 + n_y^2 + n_z^2\right)\frac{\pi^2\hbar^2}{2ma^2} \qquad \text{(I)}$$

The corresponding eigenfunction in three dimensions is given by

$$\psi = \psi_x\psi_y\psi_z = \left(\sqrt{\frac{2}{a}}\right)^3 \sin\frac{n_x\pi x}{a}\sin\frac{n_y\pi y}{a}\sin\frac{n_z\pi z}{a} \qquad \text{(II)}$$

Example 5

Find the probability that a particle trapped in a box L wide can be found between 0.45L and 0.55L for the ground and first excited states.

Solution:

Probability = $\int_{0.45L}^{0.55L}|\psi_n|^2\,dx = \frac{2}{L}\int_{0.45L}^{0.55L} sin^2\frac{n\pi x}{L}\,dx = \left[\frac{x}{L} - \frac{1}{2n\pi}\sin\frac{2n\pi x}{L}\right]_{0.45\,L}^{0.55\,L}$

For the ground state, i.e., n = 1, Probability = 0.198 = 19.8%

For the first excited state, i.e., n = 2, Probability = 0.0065 = 0.65%

Example 6

Find the expectation value for the position <x> and momentum <p> of a particle trapped in a box of width L.

Solution:

$\langle x\rangle = \int_{-\infty}^{\infty} x|\psi|^2 dx = \frac{2}{L}\int_0^L x sin^2\frac{n\pi x}{L}\,dx = \frac{2}{L}\left[\frac{x^2}{4} - \frac{x\sin(2n\pi x/L)}{(4n\pi/L)} - \frac{\cos(2n\pi x/L)}{8(n\pi/L)^2}\right]_0^L$

Putting $\sin n\pi = 0, \cos 2n\pi = 1\ and\ cos0 = 1$, we obtain

$$\langle x\rangle = \frac{2}{L}\left(\frac{L^2}{4}\right) = \frac{L}{2}$$

$\psi^* = \psi = \sqrt{\frac{2}{L}}\,sin\frac{n\pi x}{L}$; $\frac{d\psi}{dx} = \sqrt{\frac{2}{L}}\frac{n\pi}{L}\cos\frac{n\pi x}{L}$

$$\langle p\rangle = \int_{-\infty}^{\infty}\psi^*\hat{p}\psi\,dx = \int_{-\infty}^{\infty}\psi^*\left(-i\hbar\frac{d}{dx}\right)\psi\,dx = -i\hbar\,\frac{2}{L}\frac{n\pi}{L}\int_0^L sin\frac{n\pi x}{L}\cos\frac{n\pi x}{L}\,dx$$

Now, $\int \sin ax\cos ax\,dx = \frac{1}{2a}sin^2 ax$

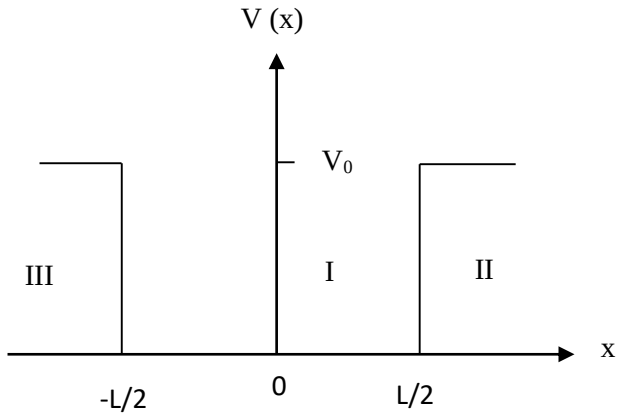
Substituting a = $\frac{n\pi}{L}$, we get $\langle p\rangle = 0$

Particle trapped in a finite potential well or a square potential well with finite barriers

Consider a particle of mass m and energy E in a finite potential well with finite barriers.

The finite potential well is an extension of the infinite potential well. The main difference between these two systems is that now the particle has a non-zero probability of finding itself outside the well, although its kinetic energy is less than that required, according to classical mechanics, for scaling the potential. This type of problem is more realistic, but more difficult to solve due to the yielding of transcendental equations. The particle is again confined to a box, but one which has finite, not infinite, potential walls.

The Finite Potential Well



Consider a particle in the potential

$$V(x) = \begin{cases} V_0 & x < -\frac{L}{2} \\ 0 & -\frac{L}{2} \le x \le \frac{L}{2} \\ V_0 & x > \frac{L}{2} \end{cases}$$

Here,

$$V_0 > 0$$

Let E be the energy of the particle incident on

this potential well of height V_o.

When the energy of the particle is less than that of the potential, i.e., when $E < V_0$, it gives rise to a bound state. The energy spectrum of the bound states is discrete, unlike free particles, which have a continuous spectrum.

On the other hand, when the energy of the particle is greater than that of the potential, i.e., when $E > V_0$, it gives rise to an unbound state. The unbound states have a continuous energy spectrum.

Solving the time independent Schrodinger equation in all the three regions gives the energy and wave-function of the particle.

$-\frac{L}{2} \le x \le \frac{L}{2}$ (Region I):

The time independent Schrodinger equation in this region is

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) = E\psi(x) \;\Rightarrow\; \psi''(x) = -k^2\psi(x)$$

Here, $\boxed{k^2 = \frac{2mE}{\hbar^2} > 0}$

The general solution of the above equation is

$$\psi_I(x) = A\sin kx + B\cos kx,$$

where A and B are arbitrary constants.

$x > \frac{L}{2}$ (Region II):

Consider that the total energy E of the particle in region II is less than V_0 i.e., $E < V_0$. This the classically forbidden region since the particle would reflect off from the potential wall completely. The same system in quantum mechanics gives a non-zero probability that the particle will be transmitted into this region. This is a wave phenomenon, but in quantum mechanics particles exhibit wave-like properties.

The time independent Schrodinger equation in this region is

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V_0\psi(x) = E\psi(x)$$

or

$$\psi''(x) = \alpha^2\psi(x)$$

where

$$\boxed{\alpha^2 = \frac{2m(V_0 - E)}{\hbar^2} > 0}$$

The general solution of the above equation is

$$\psi_{II}(x) = Ce^{-\alpha x} + De^{\alpha x}$$

where C and D are arbitrary constants.

For positive values of x, the second term of the above equation ($De^{\alpha x}$) will increase exponentially and the solution $\psi_{II}(x)$ will not be square integrable. Hence, the value of D must be zero for the solution $\psi_{II}(x)$ to be square integrable.

$x < -\frac{L}{2}$ (Region III):

The time independent Schrodinger equation in this region is

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V_0\psi(x) = E\psi(x)$$

or

$$\psi''(x) = \alpha^2\psi(x)$$

where

$$\boxed{\alpha^2 = \frac{2m(V_0 - E)}{\hbar^2} > 0}$$

The general solution of the above equation is

$$\psi_{III}(x) = Fe^{-\alpha x} + Ge^{\alpha x}$$

For large negative values of x, the first term of the above equation ($Fe^{-\alpha x}$) will increase exponentially and the solution $\psi_{III}(x)$ will not be square integrable. Hence, the value of F must be zero for the solution $\psi_{III}(x)$ to be square integrable.

Since, the potential V(x) is symmetric about x = 0, the solutions can be both symmetric (having even-parity) and anti-symmetric (having odd-parity).

Considering even-parity solutions in region I, we get

$$\psi_I(x) = B\cos kx$$

Now, we apply the following two boundary conditions that a well-behaved wave-function must satisfy.

1. ψ continuous at $x = L/2$:

$$\psi_I(L/2) = \psi_{II}(L/2)$$

$$B\cos(kL/2) = C\exp(-\alpha L/2) \tag{1}$$

2. ψ' continuous at $x = L/2$

$$\psi_I'(L/2) = \psi_{II}'(L/2)$$

$$-Bk\sin(kL/2) = -C\alpha\exp(-\alpha L/2) \tag{2}$$

Dividing eqn. (2) by eqn. (1), we get,

$$\boxed{\tan\frac{kL}{2} = \frac{\alpha}{k}} \tag{3}$$

Similarly, considering odd-parity solutions in region I, we get

$$\psi_I(x) = B\sin kx$$

and then applying the boundary conditions,

1. ψ continuous at $x = L/2$:

$$\psi_I(L/2) = \psi_{II}(L/2)$$

$$B\sin(kL/2) = C\exp(-\alpha L/2) \tag{4}$$

2. ψ' continuous at $x = L/2$

$$\psi_I'(L/2) = \psi_{II}'(L/2)$$

$$Bk\cos(kL/2) = -C\alpha\exp(-\alpha L/2) \tag{5}$$

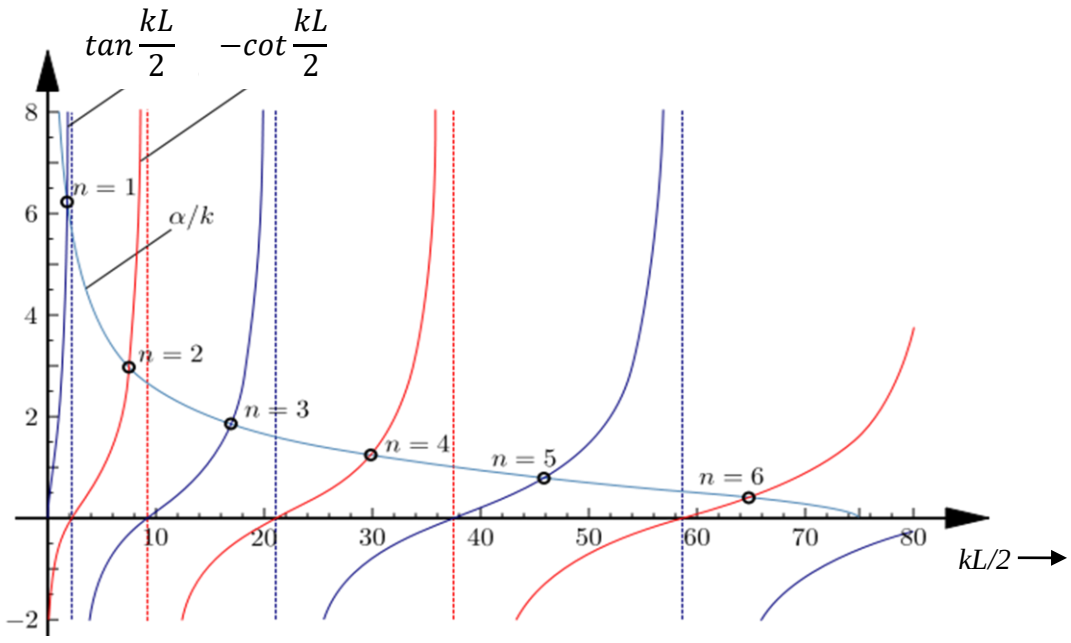
Dividing eqn. (5) by eqn. (4) we get,

$$\cot \frac{kL}{2} = -\frac{\alpha}{k}$$

(6)

The equations (3) and (6) are transcendental equations and their solutions can be obtained graphically by determining the points where the curve y = tan (*kL*/2) intersects the curve

y = α/*k* *or* where the curve y = - cot (*kL*/2) intersects the curve y = α/*k*

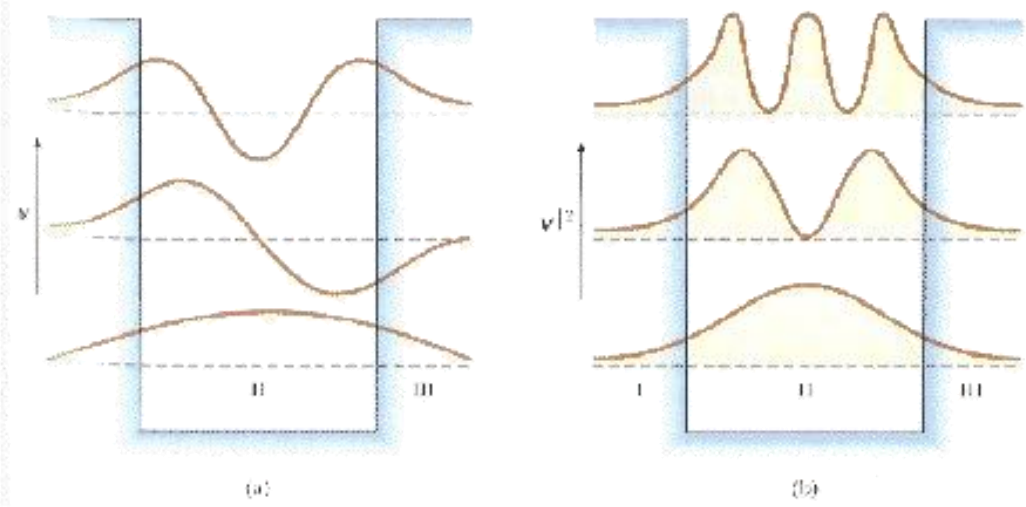


The points of intersection determine the values of k and the energy of the particle is

$E = \frac{\hbar^2 k_n^2}{2m}$ where $k_n = \frac{n\pi}{L}$

For even-parity or symmetric states, the values of n = 1, 3, 5, ...

while for odd-parity or anti-symmetric states, the values of n = 2, 4, 6, ...



The lowest energy state (ground state) is always above the bottom of the finite potential well and always corresponds to a symmetric state.

The first three wave functions and their energies are shown in the above figure. Note the oscillatory behaviour of the wave inside the well, and the exponential decay of the wave amplitude into the classically disallowed regions (where the kinetic energy of the particle is zero). This new effect of barrier penetration is a purely quantum phenomenon. There is a smooth matching of the exponential part of the wave to the oscillatory part. The corresponding probability density diagrams indicate the probability of finding the particle in the classically non-allowed regions.

The phenomenon of barrier penetration by a particle is termed as quantum tunnelling. Hence, quantum particles can be found in regions where a classical particle could never go, since, it would have negative kinetic energy. But in quantum theory, too, the eigenvalues of kinetic energy cannot be negative. How, then, can a quantum particle "tunnel"?

The apparent paradox is resolved by noting that the wave-function of a tunnelling particle only partly overlaps the forbidden region. There is no wave function that represents a particle restricted to a region where its potential energy is larger than its total energy.

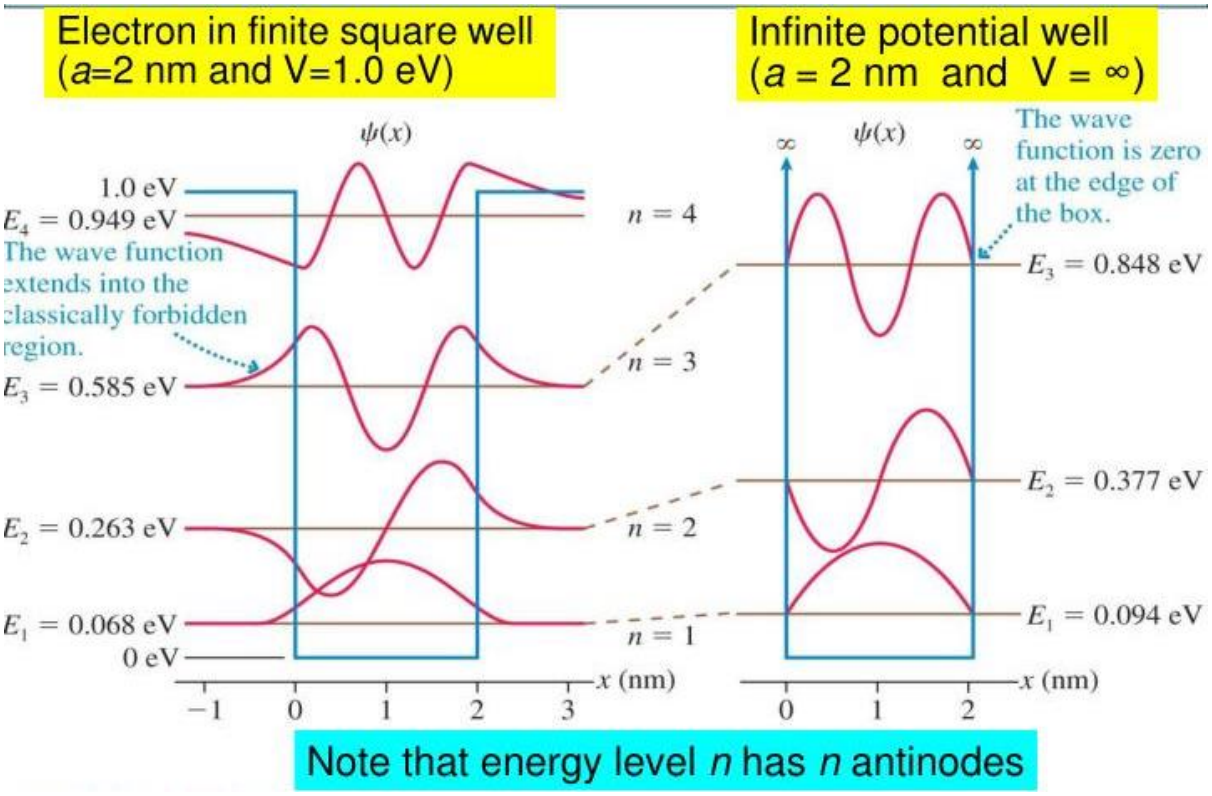
The wavefunction of the tunneling particle decreases exponentially in the barrier. A measure of the depth of tunnelling is determined by the penetration depth (λ). It is related to α where

$$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

by the relation $\lambda = 1/\alpha$. At the distance equal to the penetration depth, the wave-function is reduced by a factor of 1/e.

The tunneling probability is strongly dependent on the width of the barrier, the mass of the particle, and the quantity V-E. For instance, the ratio of tunneling probability for protons to electrons is around a factor of 10^-91. So electrons are much more likely(!) than protons at the same energy to tunnel through the same barrier.

Comparison between infinite and finite potential well



1. The wave-function $\psi(x)$ is confined to the infinite well while $\psi(x)$ spreads out beyond the finite well.
2. The energy levels in a finite potential well are lower than the infinite well as the wavefunction spreads out (by penetrating the classically forbidden region) which reduces its kinetic energy.
3. There are no unbound states in an infinite potential well while there are unbound states in a finite potential well when $E > V_0$.